

Supporting Information for

# Modeling Dynamic Risk Assessment for Autonomous-Systems Conflict Resolution: Constraint-Aware and Distribution-Sensitive Decision-Making Framework

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## Abstract

This technical research artifact provides the supporting information, detailed architecture, mathematical formulations, implementation parameters, and canonical experimental reproducibility notes for the Dynamic Risk Assessment (DRA) framework. It documents the exact execution pipeline, state machine transitions, control adapter synchronization, and data provenance chain supporting the manuscript.

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## 1 Artifact Overview & Provenance

- **Project Title:** Dynamic Risk Assessment (DRA) for Autonomous Vehicles
- **Canonical Code Version:** DRA\_v4.3.2.Resume\_State\_Fix.py
- **Canonical Execution Timestamp:** 20260813\_143431
- **Primary Data Logs:**  
timestep\_log\_20260813\_143431.csv,  
episode\_results\_20260813\_143431.csv
- **Environment Log File:** DRA\_Debug\_v4.3.2.2.txt

## 2 System Architecture & Mathematical Formulation

The DRA framework evaluates candidate maneuvers in a road-aligned local coordinate system using planar rotation:

$$\begin{bmatrix} x_r \\ y_r \end{bmatrix} = \mathbf{R}^T \left( \begin{bmatrix} x \\ y \end{bmatrix} - \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} \right), \quad \mathbf{R} = \begin{bmatrix} \cos \psi & -\sin \psi \\ \sin \psi & \cos \psi \end{bmatrix} \quad (1)$$

## 2.1 Risk Fields & Distributional Terms

Pedestrian risk field is computed as:

$$R_{\text{ped}}(k) = R_{\text{ped},0} \exp\left(-\frac{d_{\text{ped}}^2(k)}{2\sigma_{\text{ped}}^2}\right) \quad (2)$$

To capture ethical risk allocation beyond aggregate scalar costs, DRA incorporates distribution-sensitive terms:

$$R_{\text{maximin}} = \max(\bar{R}_{\text{ped}}, \bar{R}_{\text{oncoming}}) \quad (3)$$

$$R_{\text{equality}} = |\bar{R}_{\text{ped}} - \bar{R}_{\text{oncoming}}| \quad (4)$$

## 2.2 State Machine Transitions

The interaction progresses through four explicit operational states:

NORMAL  $\longrightarrow$  APPROACH\_STOP  $\longrightarrow$  WAIT  $\longrightarrow$  RESUME  $\longrightarrow$  NORMAL

## 3 Canonical Quantitative Summary

Table 1: Canonical Evaluation Metrics Across Evaluated Planners

| Metric                       | TTC Baseline | Utilitarian | Full DRA      |
|------------------------------|--------------|-------------|---------------|
| Successful Episodes          | 0/2          | 0/2         | <b>2/2</b>    |
| Pedestrian Collisions        | 2/2          | 0/2         | <b>0/2</b>    |
| Oncoming Lane Duration (s)   | 0.000        | 6.925       | <b>0.000</b>  |
| Min Pedestrian Clearance (m) | 0.705        | 5.648       | <b>6.915</b>  |
| Integrated Pedestrian Risk   | 2026.31      | 203.30      | <b>180.02</b> |
| Peak Pedestrian Risk         | 340.03       | 102.95      | <b>24.56</b>  |
| Mean Planning Time (ms)      | 0.007        | 0.050       | 59.63         |

## 4 Video Demonstration Index

The following media files accompany this record on Zenodo:

### A. Canonical Runs:

- DRA\_Qualitative\_Run\_Evasive\_Maneuver.mp4
- DRA\_Qualitative\_Run\_Stop\_Wait\_Resume.mp4

### B. Historical Development Evidence (Non-canonical benchmark):

- Historical\_Development\_Pavement\_Encroachment\_Scenario.mp4
- Historical\_Development\_Aggressive\_Initial\_Velocity\_DRA\_LowSpeed\_Interaction.mp4

## 5 Reproducibility Notes

To reproduce the canonical run:

1. Launch CARLA Simulator (0.9.x) on Town03.
2. Execute `python src/DRA_v4.3.2.Resume_State_Fix.py --seed 1000 --scenario S1`.
3. Logs will generate directly under `results/canonical/`.